WORKED SOLUTION

STEP I 2003, Question 9

Mechanics • Projectile Through a Point

STEP I 2003, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

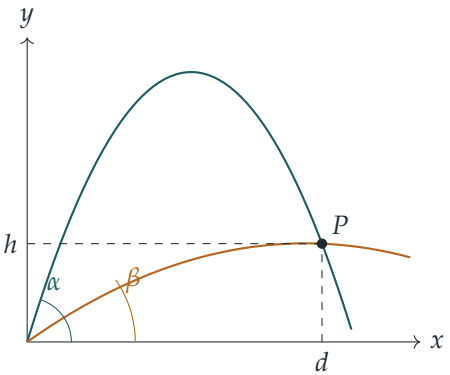
step.maths.org/questions/03-s1-q9

The key idea

Write down the horizontal and vertical displacements, eliminate the time, and replace $\sec^2\theta$ by $1+\tan^2\theta$: the trajectory equation becomes a quadratic in $T=\tan\theta$. After that everything is about this quadratic. Two angles means two distinct real roots, so look at the discriminant. And $\alpha+\beta$ is controlled by the sum and product of the roots, through the formula for $\tan(\alpha+\beta)$.

Solution

We model the particle as moving freely under gravity alone (no air resistance), with the origin at the point of projection, x horizontal towards P and y vertically upwards. So P is the point (d,h) with $d>0$. We take $h>0$, meaning that P is above the point of projection; we come back to $h\leq 0$ at the end.



The quadratic in T

The horizontal velocity is constant, $V\cos\theta$, and the vertical motion has constant acceleration $-g$, so at time t

$$x = Vt\cos\theta, \qquad y = Vt\sin\theta - \tfrac{1}{2}gt^2.$$

The particle reaches $x = d$ at time $t = d/(V\cos\theta)$ (here $\cos\theta > 0$, as the particle moves towards P). Substituting this into y and setting $y = h$,

$$h = d\tan\theta - \frac{gd^2}{2V^2\cos^2\theta} = dT - \frac{gd^2}{2V^2}(1+T^2),$$

using $\sec^2\theta = 1 + \tan^2\theta$. Now multiply through by $\dfrac{2V^2}{gd^2} = \dfrac{2k}{d}$:

$$\frac{2kh}{d} = 2kT - (1+T^2), \qquad \text{that is,} \qquad T^2 - 2kT + \frac{2kh}{d} + 1 = 0,$$

as required.

Two distinct angles

For a given V (and so a given k), the possible angles of projection are the $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ whose tangent is a root of this quadratic. Since tan is strictly increasing on that interval, distinct real roots T give distinct angles. So we need the quadratic to have two distinct real roots, which happens exactly when its discriminant is positive:

$$4k^2 - 4\Big(\frac{2kh}{d} + 1\Big) > 0 \iff k^2d^2 - 2h(kd) - d^2 > 0 \iff (kd-h)^2 > h^2 + d^2,$$

multiplying by $d^2 > 0$ and then completing the square in kd.

Ask yourself Why does the given condition imply $(kd-h)^2 > h^2 + d^2$, when squaring inequalities can go wrong?

If $kd > h + \sqrt{h^2+d^2}$, then $kd - h > \sqrt{h^2+d^2} > 0$. Both sides are positive, so squaring keeps the inequality: $(kd-h)^2 > h^2 + d^2$. So the discriminant is positive and there are two distinct roots T, hence two distinct angles of projection.

(The converse holds too. The other way to have $(kd-h)^2 > h^2+d^2$ is $kd < h - \sqrt{h^2+d^2}$, but the right-hand side is negative since $\sqrt{h^2+d^2} > |h|$, while $kd = V^2/g > 0$. So two angles are possible *exactly* when $kd > h + \sqrt{h^2+d^2}$.)

Both angles are genuinely *above* the horizontal. The roots have sum $2k > 0$ and product $1 + 2kh/d > 0$ (as $h > 0$), so both are positive.

The sum of the angles

Let $\tan\alpha$ and $\tan\beta$ be the two roots. From the quadratic,

$$\tan\alpha + \tan\beta = 2k, \qquad \tan\alpha\tan\beta = 1 + \frac{2kh}{d}.$$

The product is not 1 (since $kh/d \neq 0$), so $\alpha + \beta \neq \frac{\pi}{2}$ and the compound-angle formula applies:

$$\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{2k}{-2kh/d} = -\frac{d}{h}.$$

Note that k has cancelled: the sum of the angles doesn't depend on the speed.

Ask yourself $\tan(\alpha+\beta) = -d/h$ has infinitely many solutions. Which one is $\alpha+\beta$?

We showed $\alpha, \beta \in (0, \frac{\pi}{2})$, so $0 < \alpha+\beta < \pi$. Moreover $\tan(\alpha+\beta) < 0$, so $\alpha+\beta$ lies in $(\frac{\pi}{2}, \pi)$. The one angle in $(\frac{\pi}{2}, \pi)$ with tangent $-d/h$ is $\pi - \arctan(d/h)$, because $\arctan(d/h) \in (0, \frac{\pi}{2})$ and $\tan(\pi - u) = -\tan u$. Therefore

$$\alpha + \beta = \pi - \arctan\left(\frac{d}{h}\right),$$

as required.

There is a nice way to see this. $\arctan(d/h)$ is the angle between OP and the upward vertical, so $\pi - \arctan(d/h) = \frac{\pi}{2} + \gamma$, where γ is the angle of elevation of P. Then $\frac{1}{2}(\alpha+\beta) = \frac{\pi}{4} + \frac{\gamma}{2}$: the two directions of projection are symmetric about the line bisecting the angle between OP and the vertical.

Checks

- If $h = 0$ (the target is on the same level) the formula gives $\alpha + \beta = \pi - \frac{\pi}{2} = \frac{\pi}{2}$: the familiar complementary pair of angles with equal range.
- Take $d = 3$, $h = 4$, $k = 4$. Then $kd = 12 > 4 + 5$, and the roots are $T = 4 \pm \sqrt{13/3}$, giving $\alpha \approx 1.4078$ and $\beta \approx 1.0903$. So $\alpha + \beta \approx 2.4981$, which equals $\pi - \arctan(0.75) \approx 2.4981$.
- A numerical simulation of the flight at each of the two angles, for thousands of random d, h and k, passes through (d, h) and confirms both the condition and the sum of the angles.

Common slips

- Forgetting that $\sec^2\theta = 1 + \tan^2\theta$, and so never reaching a quadratic in T.
- Writing $\alpha + \beta = \arctan(-d/h) = -\arctan(d/h)$, which is negative. The inverse tangent only returns angles in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so you must use the range of $\alpha+\beta$ to pick the correct branch.
- Taking $h < 0$ (a target below the point of projection) without care. Then one root may be negative, and the correct general statement is $\alpha + \beta = \frac{\pi}{2} + \arctan(h/d)$. This agrees with the question's form only when $h > 0$.

Beyond STEP

The boundary case $kd = h + \sqrt{h^2 + d^2}$, where the two angles merge, traces out the *envelope* of all trajectories at speed V: the "parabola of safety" $y = \dfrac{V^2}{2g} - \dfrac{gx^2}{2V^2}$. Points inside it can be hit in two ways, points on it in one way, and points outside not at all. Envelopes of families of curves, found by solving an equation and its derivative with respect to the parameter at the same time, come up again in differential equations and optics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops projectile and constrained motion with this kind of care.

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